

# LLM-ERP Product Overview (English) — v3.0 Conversational ERP

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One-page snapshot → Full buying guide for decision makers Audience: Owners / Procurement managers / IT directors / System integrators / Consultants

⚡ v3.0 Strategic Pivot Notice (2026-05-15) Pitch updated: "Conversational ERP for 50-100 person factories — open Chrome, type one sentence to Chat, do CRUD on all documents at 1/10 the cost of SAP."

Deprecated sections (moved to Phase 7 pending customer feedback):

- "Outsource LINE Coordination" chapter
- Mobile App / LINE Bot marketing content



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# 1. One-Page Snapshot

| Item                | Detail                                                                       |
|---------------------|------------------------------------------------------------------------------|
| Product             | LLM-ERP (AI-Native ERP for SMB Manufacturers)                                |
| Positioning         | The <b>LINE-Native ERP</b> for 50-100 person factories                       |
| Core Promise        | The owner runs the factory from LINE; live in 2 weeks, productive in 2 hours |
| Three Weapons       | ① Natural language ops ② Mobile-first ③ Outsource LINE collaboration         |
| Tech Foundation     | FastAPI + PostgreSQL/SQLite + Multi-Agent LLM + React/Expo                   |
| Deployment          | One-click Docker, self-host / cloud / MESH multi-factory                     |
| Language            | Traditional Chinese + English (full i18n)                                    |
| Devices             | Desktop browser + iOS app + Android app + LINE Bot                           |
| Onboarding Time     | < 2 weeks (vs SAP 6-18 months)                                               |
| Training / employee | < 2 hours (vs traditional 1-3 months)                                        |
| Typical Price       | NT\$ 300K-500K / year (vs SAP NT\$ 2M+)                                      |

**In one sentence:** We collapse "6-month rollout, 3-month training" pain into "2 weeks live, 2 hours productive" by letting **AI translate the system for users**.

## 2. Product Positioning

### 2.1 Ideal Customer Profile (ICP)

| Dimension        | Range                                                             |
|------------------|-------------------------------------------------------------------|
| Headcount        | 50-100 people (often 10-20 in-house + outsource chain)            |
| Industry         | Metal machining / Plastic injection / PCB / Food / Textile dyeing |
| Revenue          | NT\$ 100M-1B / year                                               |
| Digital maturity | Low-to-mid (still on Excel + LINE groups)                         |
| IT budget        | NT\$ 300K-1M / year                                               |
| Owner role       | Reads numbers personally, no dedicated IT department              |

### 2.2 Five Typical Users (Persona)

| Role                   | Daily Scene                                                            | Primary Device |
|------------------------|------------------------------------------------------------------------|----------------|
| Owner (Mr. Wang)       | 06:30 asks LINE "today's status" → gets one-page dashboard             | LINE           |
| Salesperson (Chen)     | In customer's office, looks up M6 stock on phone, quotes in 3s         | Phone          |
| Plant Manager (Lin)    | On factory floor, gets push notification "WO-2026-100 stuck"           | Phone + LINE   |
| Buyer/Warehouse (Ling) | Scans QR on incoming carton, inventory auto-increments                 | Phone (scan)   |
| Outsource Master (Wu)  | LINE receives drawing, snaps photo to report progress, no registration | LINE           |

2.3 Pains We Solve

| Customer Voice                                    | Our Solution                                          |
|---------------------------------------------------|-------------------------------------------------------|
| "ERP is too expensive, we can't afford SAP"       | Open-source core + NT\$ 300K/year subscription        |
| "6-month rollout? Who can endure that?"           | Docker one-click + natural language → live in 2 weeks |
| "Training employees is hell"                      | LINE + AI chat interface → no system to learn         |
| "Inventory out of sync, sales quote wrong stock"  | Live DB + SSE push + 3-second mobile query            |
| "Outsource shops can't see our progress"          | LINE QR + Bot reporting → no registration needed      |
| "Boss wants live numbers, not month-end reports"  | LINE Bot real-time dashboard                          |
| "We only notice missing parts when it's too late" | EventBus proactive alerts                             |

3. 12 Core Modules

Each domain has a complete RESTful API + desktop UI + mobile UI + natural-language interface.

| #  | Domain        | Key Features                                                       | Primary Users        |
|----|---------------|--------------------------------------------------------------------|----------------------|
| 1  | Inventory     | Parts / BOM / Transactions / Safety stock alerts / Multi-warehouse | Warehouse + Sales    |
| 2  | Purchase      | Supplier tiering (T1-T3) / PO / Receiving / 3-way match / Auto AP  | Buyer                |
| 3  | Sales         | Customer / Quote / SO / Shipment / Auto AR                         | Sales + Owner        |
| 4  | Production    | Work orders / BOM explode / Dispatch / Reporting / OEE             | Plant Mgr + Operator |
| 5  | Quality       | Inspection / Non-conformance / CAPA / Customer complaints          | QC + QA              |
| 6  | Warehouse     | Multi-warehouse / Barcode count / Transfer / Auto replenishment    | Warehouse            |
| 7  | MPS/MRP       | Master schedule / Material req planning / Simplified for SMB       | Planner + Plant Mgr  |
| 8  | Accounting    | AR / AP / Month-end close / Auto journals                          | Finance              |
| 9  | CRM           | Lead → Opportunity → Customer funnel + Visit log                   | Sales                |
| 10 | HR            | Employee / Department / Role / Training                            | HR + Manager         |
| 11 | AI Governance | DecisionLog (auditable AI calls) / Model switching                 | IT + Audit           |
| 12 | Permission    | RBAC 5 layers / 109 perms / 11 roles / Row-Level filter            | IT + Owner           |

3.1 Cross-Domain Intelligence

- **Multi-Agent AI:** 10 specialist agents (Inventory / Sales / Production / Quality / Purchase / Planning / Warehouse / Accounting / CRM / General) + 26 tools
- **EventBus:** 16 Constraint Rules auto-fire across domains (e.g. "PO approved" → notify supplier + warehouse)
- **MESH Multi-Factory:** Cross-factory inventory aggregation / VMI (raw data stays in each factory)
- **SSE Live Push:** Inventory changes, WO status, AI alerts reach users in sub-second

## 4. Key Differentiators

### 4.1 Natural Language Operation (Killer Feature #1)

```
Owner types: "Today's factory status"
↓
IntentClassifier → ProductionAgent
↓
Tool call: query_today_summary
↓
Reply: 12 WOs today (8 in-progress), 3 items below safety stock,
       5 shipments on track 95%
```

**Contrast:** Traditional ERP requires 7 menu clicks + 3 reports for the same info.

### 4.2 Mobile-First Design

| Principle       | Realization                               |
|-----------------|-------------------------------------------|
| Mobile-first    | 5 tabs, big buttons, minimal typing       |
| 3-tap rule      | Any function $\leq 3$ taps                |
| Offline degrade | AsyncStorage caches token + basic data    |
| Large fonts     | Readable outdoor by owner / plant manager |
| One-hand reach  | Key actions in thumb zone                 |

### 4.3 Outsource LINE Collaboration (Industry First)

```
HQ → Via LINE Bot sends drawing + process + one-time QR token
↓
Outsource master (no registration in any system) → opens LINE → scans QR
↓
Reports "80% complete" + photo
↓
HQ work order auto-updates / EventBus triggers QC notification
```

**Industry pain:** Traditional ERPs require outsource shops to register / learn new software. Masters refuse. **Our solution:** Outsource shops **don't register** at all. If they use LINE, they can use this.

## 4.4 MESH Multi-Factory Data Sovereignty

HQ — WireGuard VPN —

├ Main Factory (A) :8001

├ Plating Outsource (B) :8002

├ Inspection Outsource (C) :8003

└ Nth Outsource

HQ asks "Total M6 across all factories?" →

• HQ parallel-fans to N factories /api/factory/mesh/query

• Each factory returns only its own aggregated sum (no raw rows)

• HQ totals: 3000 + 1500 + 800 = 5300

★ HQ NEVER sees "which batch, who bought it"

★ Integration test 5/5 PASS – not a stub, actually works

# 5. Technical Spec

## 5.1 System Requirements

| Resource      | Min                   | Recommended                 |
|---------------|-----------------------|-----------------------------|
| CPU           | 2 core                | 4 core                      |
| RAM           | 4 GB                  | 8 GB                        |
| Disk          | 20 GB SSD             | 100 GB SSD                  |
| OS            | Win 10 / Mac / Linux  | Linux Server                |
| Docker        | 20.10+                | 24+                         |
| Network       | 100 Mbps LAN          | 1 Gbps                      |
| Public access | HTTPS (LINE Bot only) | Cloudflare Tunnel free tier |

5.2 Backend Stack

| Layer       | Tech                                            |
|-------------|-------------------------------------------------|
| Web         | FastAPI (Python 3.12+) async                    |
| ORM         | SQLAlchemy 2.0 async                            |
| Database    | PostgreSQL 15+ / SQLite (sufficient for SMB)    |
| Migration   | Alembic                                         |
| Auth        | JWT (HS256) + RBAC 5-layer                      |
| Real-time   | Server-Sent Events (SSE)                        |
| LLM Adapter | Claude / DeepSeek / OpenAI / Ollama (pluggable) |
| Container   | Docker Compose                                  |

5.3 Frontend Stack

| Surface    | Tech                                                     |
|------------|----------------------------------------------------------|
| Desktop UI | React 18 + Vite + Tailwind + Zustand + TypeScript strict |
| Mobile App | Expo SDK 51 + React Native + expo-router                 |
| LINE Bot   | LINE Messaging API + Webhook (planned)                   |
| i18n       | Custom lightweight (zh-TW + en)                          |



5.4 Performance Targets

| Operation                    | Target latency | Measured              |
|------------------------------|----------------|-----------------------|
| Health check                 | < 100 ms       | ~20 ms                |
| Login                        | < 500 ms       | ~200 ms               |
| List query (<1000 rows)      | < 1 s          | ~300 ms               |
| AI chat (simple)             | < 5 s          | 2-5 s (LLM-dependent) |
| AI chat (multi-step tool)    | < 30 s         | 6-15 s                |
| MESH aggregate (3 factories) | < 3 s          | ~500 ms               |
| SSE push latency             | < 2 s          | < 1 s                 |

5.5 Scale Limits (Single Instance)

| Dimension        | Limit             |
|------------------|-------------------|
| Concurrent users | 100-200           |
| Requests / sec   | 200-500 RPS       |
| Inventory parts  | 100,000           |
| Work orders      | 100,000 / year    |
| Conversation log | 1,000,000 entries |
| MESH nodes       | 10-20             |

Beyond this scale, recommend factory-distributed + MESH aggregation.

## 6. Deployment Options

### 6.1 Four Modes

Mode A: Single-machine Docker (simplest, 50-person factory)

- └ HQ backend + Desktop UI + DB all on one box
- └ One-click install.bat / install.sh

Mode B: Managed cloud (for factories without IT)

- └ We host on GCP/AWS/Azure
- └ Customer sees only https://your-company.llm-erp.app
- └ Monthly fee covers host + backup + upgrade

Mode C: Customer self-host (data sovereignty)

- └ Customer buys VM / on-prem server
- └ We provide Docker image + deployment guide
- └ Customer's IT maintains

Mode D: MESH multi-factory (group / multi-site owners)

- └ HQ + N Factory Nodes
- └ WireGuard VPN mesh
- └ Each factory's data stays on-site
- └ HQ sees aggregates only

### 6.2 Install Time

| Mode              | First install      | Customer does                    |
|-------------------|--------------------|----------------------------------|
| A — Single Docker | 10 min             | Double-click install.bat         |
| B — Managed cloud | Next day live      | Fill form + pay                  |
| C — Self-host     | 0.5-1 day          | Remote pairing with our engineer |
| D — MESH          | 1-3 days / factory | Install per factory + setup VPN  |

## 7. Integration Capabilities

### 7.1 Outbound Integrations

| Target        | Method                      | Purpose                               |
|---------------|-----------------------------|---------------------------------------|
| LINE          | Messaging API + Bot Webhook | Owner dashboard / outsource reporting |
| Excel         | CSV / xlsx export           | Owners prefer Excel                   |
| Email         | SMTP (SendGrid-compatible)  | PO PDF / reports                      |
| 3rd-party ERP | REST API + Webhook          | Upstream customer / group             |
| Customs / Gov | Reserved interface          | Export / customs filing               |

### 7.2 LLM Providers (Pluggable)

| Provider                        | Best For                              | Monthly Cost (50p)         |
|---------------------------------|---------------------------------------|----------------------------|
| Anthropic Claude                | International brand orders, strictest | NT\$ 3,000-8,000           |
| OpenAI GPT-4o                   | Industry standard                     | NT\$ 2,500-6,000           |
| DeepSeek                        | Chinese context, best ROI             | NT\$ 800-2,000             |
| Ollama local (gemma3 / qwen2.5) | Zero data egress                      | NT\$ 0 (one-time hardware) |

Switching LLM = edit one `.env` line + restart. See [LLM Benchmark Report](#).

## 8. Security & Compliance

### 8.1 Security Architecture

| Layer         | Mechanism                                                           |
|---------------|---------------------------------------------------------------------|
| AuthN         | JWT (HS256) + bcrypt password hash                                  |
| AuthZ         | RBAC 5-layer (Tenant→User→Role→Permission→Row-Level)                |
| Audit         | AuditMiddleware logs all requests + DecisionLog (AI calls)          |
| Transport     | HTTPS (nginx / Cloudflare Tunnel)                                   |
| Multi-factory | WireGuard VPN + factory VLAN                                        |
| Secrets       | <code>.env</code> + HashiCorp Vault recommended (enterprise)        |
| Demo bypass   | Active only when JWT_SECRET is default; auto-disables in production |

### 8.2 Data Sovereignty

- MESH multi-factory mode: Raw rows physically stay in each factory's SQLite/PostgreSQL
- HQ can only call `/api/factory/aggregate` for aggregated numbers
- Integration test verified: aggregate response contains **none** of `created_at` / `qty_available` / `items` / `rows`

### 8.3 Compliance Readiness

| Standard                | Status                                               |
|-------------------------|------------------------------------------------------|
| GDPR (right to erasure) | API endpoint reserved, customer enables              |
| ISO 27001               | Documentation 70% complete, requires 3rd-party audit |
| China PIPL              | Can deploy to China DC (DeepSeek/Ollama)             |
| HIPAA                   | Healthcare requires custom encryption                |

## 9. Onboarding Process & Timeline

### 9.1 Standard Rollout (2-Week Model)

|           |                                                |
|-----------|------------------------------------------------|
| Day 1-2   | Contract + Requirements interview              |
| Day 3-5   | Environment prep (server / accounts / network) |
| Day 6-7   | Docker install + industry-sample data load     |
| Day 8-9   | Customer data import (from Excel)              |
| Day 10    | Admin training (2 hours)                       |
| Day 11-12 | Department head training (5 roles × 1 hour)    |
| Day 13    | Internal pilot                                 |
| Day 14    | Go-live                                        |

### 9.2 Extended Rollout (Multi-factory MESH, 4-6 weeks)

Additional: per-factory install, WireGuard VPN setup, cross-factory permissions, aggregation tests.

### 9.3 Training Hours

| Role                 | Hours                           |
|----------------------|---------------------------------|
| Owner / Plant Mgr    | 1 hour (LINE-based)             |
| Sales / Buyer        | 2 hours (incl. mobile app)      |
| Warehouse / Operator | 3 hours (incl. barcode scan)    |
| IT Admin             | 1 day (incl. RBAC + deployment) |

Compare with industry-average 1-3 months for SAP.

## 10. Service & Support

### 10.1 Service Level Agreements

| Tier  | Response Time    | Applicable            |
|-------|------------------|-----------------------|
| Basic | Business day, 8h | Standard subscription |
| Pro   | Business day, 2h | Add-on                |
| 24/7  | Anytime, 1h      | Enterprise            |

## 10.2 Support Channels

- Internal IT / integrator
- Monthly online training
- LINE customer support group
- Emergency phone (Pro+)
- Remote TeamViewer assist (Pro+)

## 10.3 Upgrade & Maintenance

- **Minor:** Monthly, auto-built + customer clicks upgrade
  - **Major:** Quarterly, new features + training session
  - **Data migration:** Auto migration test across major versions
- 

# 11. Procurement FAQ

### Q1: Why so much cheaper than SAP / Oracle?

Because we don't charge license fees. SAP's NT\$ 2M is mostly "license" + "implementation consultant". We're open-source core + subscription (annual fee). Customers buy subscription + service, not "ownership of software".

### Q2: Isn't open-source risky?

No. LLM-ERP is a commercial project; **open-source core** lets customers: ① see the code ② leave anytime without lock-in ③ find someone else to maintain. **Our value is in service**, not in trapping customers.

### Q3: Is our data safe?

Four layers of protection:

1. **Architectural:** MESH mode keeps data in each factory
2. **Transport:** HTTPS + WireGuard VPN
3. **Storage:** bcrypt passwords + optional disk encryption
4. **Audit:** AuditMiddleware logs all + DecisionLog

Can choose Ollama local LLM for **zero data egress**.

### Q4: Can we customize?

Yes, three ways:

- **Configuration:** change settings via admin UI (free)

- **Extension:** write new domain / agent / tool (consulting fee)
- **Core modification:** open source so you can fork (manage your own fork)

## Q5: What if your company goes under?

Open-source core continues to work; data is in customer's hands. Any Python engineer can maintain it (widespread stack). MESH mode means factory data is fully self-owned, even safer.

## Q6: Payment terms?

- Annual (standard)
- Quarterly (+10%)
- One-time license (self-host mode only,  $\approx 3 \times$  annual)

## Q7: Can we try first?

Yes. **Demo Mode 14-day free trial**, no signup:

1. Double-click `install.bat`
2. Click "Demo Mode" button
3. Use sample data to explore everything

## Q8: Do you have training materials?

Complete set:

- User Manual (ZH/EN)
- Installation Guide (ZH/EN)
- System Architecture (ZH/EN)
- LLM Benchmark Report (ZH/EN)
- Network Deployment Plan (ZH/EN)
- Video tutorials (planned)

All exportable to PDF ( `build_pdfs.bat` ).

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## 12. Competitive Comparison

### 12.1 Three Solutions

| Dimension                 | SAP Business One | Odoo Enterprise  | LLM-ERP             |
|---------------------------|------------------|------------------|---------------------|
| License / year            | NT\$ 1-2M        | NT\$ 300-800K    | NT\$ 300-500K       |
| Rollout time              | 6-18 months      | 2-4 months       | < 2 weeks           |
| Training / employee       | 1-3 months       | 2-4 weeks        | < 2 hours           |
| Mobile UX                 | Weak (legacy UI) | Mid (responsive) | Strong (native app) |
| AI natural language       | None             | None             | Core feature        |
| LINE Bot integration      | Custom           | Custom           | Built-in            |
| Outsource LINE collab     | None             | None             | Industry first      |
| Multi-factory sovereignty | Centralized      | Centralized      | MESH                |
| Customization             | Low (closed)     | Mid (open)       | High (open + AI)    |

### 12.2 Why Not Just Excel?

| Scenario                    | Excel              | LLM-ERP                  |
|-----------------------------|--------------------|--------------------------|
| Owner asks "today's status" | Flip 5 sheets      | One sentence reply       |
| Sales checks stock          | Open front desk PC | 3-second mobile          |
| Real-time sync              | Impossible         | SSE push                 |
| Outsource collaboration     | Chaotic LINE group | QR + auto-update         |
| Audit trail                 | None               | AuditMiddleware full log |



## 13. How to Buy

### 13.1 Purchase Process

Step 1

Download and install → Demo Mode trial (14 days)

Step 2

Fill requirements form (industry / size / customization)

Step 3

Our consultant 30-min requirements interview

Step 4

Quote (annual fee + one-time implementation)

Step 5

Sign + 50% prepayment

Step 6

Day 1 onboarding starts

Step 7

Day 14 go-live + final payment

### 13.2 Contact

| Need                   | Contact                                                                |
|------------------------|------------------------------------------------------------------------|
| Trial issues           | LINE @llmerp                                                           |
| Sales inquiry          | <a href="mailto:sales@llm-erp.example">sales@llm-erp.example</a>       |
| Tech support           | <a href="mailto:support@llm-erp.example">support@llm-erp.example</a>   |
| Integrator partnership | <a href="mailto:partners@llm-erp.example">partners@llm-erp.example</a> |

### Related Documents

- [Installation Guide \(English\)](#) / [中文](#)
- [User Manual \(English\)](#) / [中文](#)
- [System Architecture & Topology](#) / [中文](#)
- [LLM Benchmark Report](#) / [中文](#)
- [Network Deployment Plan](#) / [中文](#)
- Chinese version: `PRODUCT_OVERVIEW_ZH.md`

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